RATIONAL EXPONENTS

Multiplying roots with different indices

The radical form with different indices resists simplification, but the exponent form handles it cleanly.

01

RADICAL FORM — BEFORE

$$\sqrt{x} \cdot \sqrt[3]{x}$$

↓ Convert to exponent form — the index becomes the denominator

02

EXPONENT FORM

$$x^{1/2} \cdot x^{1/3}$$

↓ Product rule: add the exponents — over the common denominator 6: 3/6 + 2/6 = 5/6

03

SINGLE POWER

$$x^{5/6}$$

▼ WHY NOT 2/5?

↓ Convert back — the denominator 6 becomes the index

04

SINGLE RADICAL

AFTER

$$\sqrt[6]{x^5}$$